Lateral light scattering in paper - MTF simulation and measurement

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Abstract: The modulation transfer function (MTF) of 22 paper samples is computed using Monte Carlo simulations with isotropic or strongly forward single scattering. The inverse frequency at half maximum of the MTF (k_p) is found inappropriate as a single metric for the MTF since it is insensitive to the shape of the modeled and simulated MTF. The single scattering phase function has a significant impact on the shape of the MTF, leading to more lateral scattering. However, anisotropic single scattering cannot explain the larger lateral scattering observed in paper. It is argued that the directional inhomogeneity of paper requires a light scattering model with both the phase function and scattering distances being dependent on the absolute direction.

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1. Introduction

Lateral light scattering in turbid media is an important issue in a number of fields such as computer rendering [1] and optical tomography [2]. In the graphic arts, lateral light scattering in paper is important since it makes printed dots used to reproduce continuous tones appear larger. This optical dot gain was first explained by Yule and Nielsen [3], who suggested a non linear relationship between the reflectance factor of the halftone print and the fractional area coverage of the ink. Several extensions to this empirical model have been proposed over the years to account for internal reflection [4], or fluorescence [5, 6]. More advanced models are based on the substrate's point spread function (PSF), or its Fourier Transform, the modulation transfer function (MTF) [7].

To relate the lateral light scattering to the optical properties and composition of scattering media such as paper, light scattering models with lateral resolution are required. Oittinen [8] proposed a Kubelka-Munk (KM) [9] based model of the PSF of paper substrates. Arney et al. [10] showed that this model overestimates the effect of absorption on the MTF and proposed a simple model of the inverse frequency at half maximum of the MTF, denoted k_p, based only on the Kubelka-Munk scattering coefficient S. The model includes an ad-hoc offset attributed to directional inhomogeneity in paper. This directional inhomogeneity, meaning that lateral scattering is weaker than scattering along the paper thickness direction, was modeled by Mourad [11] who considered lateral fluxes in a KM framework by separating backscattered light and laterally scattered light. This leads to two scattering coefficients that are difficult to determine in practice. Assuming both coefficients to be equal, the model showed good agreement with the measurements of Arney et al. [12]. Light scattering in paper has on the other hand been shown to be strongly anisotropic, with forward scattering [13, 14]. Sormaz et al. [15] used Monte Carlo (MC) methods to simulate the reflectance from halftone prints. While the KM model approximates the light intensity within a medium with two diffuse fluxes, MC simulation solves the general radiative transfer (RT) equation including single scattering anisotropy. In a recent article [16], we presented MC simulations of the PSF of dyed paper and showed that both the absorption coefficient and the single scattering phase function have a significant impact on the PSF.

While Sormaz et al. studied the reflectance from one printed coated paper only, Arney et. al. observed for uncoated paper a larger lateral scattering than what is predicted by simulations. This calls for further analysis using MC simulations. The purpose of the present work is to compare the Arney model to MC simulations, to test the relevance of k_p as a single metric for the MTF, and to investigate whether the larger lateral scattering obtained with forward single scattering can account for the more narrow MTF measured on real paper samples.

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2. Method

The MTF of a set of commercial paper samples is simulated with the Arney model and with MC simulations. The scattering and absorption coefficients are determined from measured reflectance factor, transmittance and sample thickness, using both KM theory and MC simulations of general RT theory. For the MC simulations, the Henyey–Greenstein phase function [17] is used with asymmetry factor $g = 0$ for isotropic single scattering and $g = 0.8$ for forward single scattering. The KM model and the two MC simulations give the same reflectance factor and transmittance as the measurements. This leads to different scattering and absorption coefficients for different values of the asymmetry factor. The simulated MTFs allow for a direct comparison of the k_p obtained from the analytical Arney model, from MC simulations, and from measurements. The impact of anisotropic single scattering on the MTF is assessed by comparing the MTFs obtained with the two different values of the asymmetry factor.

2.1. Material, measurements and parameter estimation

The measurement of the MTF of paper is complicated and several mehtods have been proposed (see e.g. [18] for a review). In this work the measurements reported by Arney et al. [10] are used. Arney et al. reported measurements of the MTF of 22 paper samples at 20° incident angle using a knife edge and determined the Kubelka-Munk scattering (S) and absorption (K) coefficients of these samples in a 0°/d geometry. The set of samples includes most uncoloured paper types, from translucent (i.e. low S) papers and newsprint to coated fine papers. The reflectance factor and transmittance of these samples are calculated using KM theory, given the thickness, S, and K of each sample. For the MC simulations, the scattering (σ_s) and absorption (σ_a) coefficients are determined by optimization for the two different values of the asymmetry factor. The optimization uses the non spatially resolved DORT2002 model [19, 20] since it is much faster than MC methods and since only the total, and not the spatially resolved, reflectance is needed for the parameter estimation. The obtained σ_s and σ_a are reported in Table (1), together with the paper type, thickness and KM scattering and absorption coefficients.

2.2. Monte Carlo simulation of the edge response

The MC software Open PaperOpt [21] (available at `http://openpaperopt.sourceforge.net`) is used to compute the edge response of the the simulated substrates. A beam incident at 20° illuminates half of the 1 mm × 1 mm surface and the spatially resolved reflectance is simulated with a resolution of 10 μm. The spatially resolved reflectance (shown in Fig. 1(a)) is then averaged along the illumination edge and normalized to get the edge response (shown in Fig. 1(b)). The edge response is computed for a single layer and for a layer of infinite thickness representing an opaque pad of identical samples. For comparison with the Arney model, the paper samples are modeled as a single layer with refractive index equal to that of the surrounding thus neglecting surface reflection. To reduce noise, 10^6 wave packets were used in each simulation.

2.3. MTF calculation and characterization

The derivative of the edge response gives the line spread function (LSF, Fig. 1(b)). The simulated MTF (Fig. 1(c)) is then obtained by taking the Fourier transform of the LSF, and compared to the MTF model derived by Arney et al. [10], which for opaque samples can be written as

$$\mathrm{MTF}(w) = \frac{1}{1+5.4w/S}, \tag{1}$$

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Table 1. Sample type, thickness, and scattering and absorption coefficients determined for the three models used.

sample	type	t	KM		MC g=0		MC g=0.8	
			S	K	σ_s	σ_a	σ_s	σ_a
		µm	mm^{-1}	mm^{-1}	mm^{-1}	mm^{-1}	mm^{-1}	mm^{-1}
1	tracing paper	42	4.24	0.22	8.4	0.16	56.4	0.15
2	tracing paper	71	10.13	0	19.3	0	103.8	0
3	tracing paper	45.5	4.93	0	9.7	0	62.5	0
4	tracing paper	76.5	11.18	0	21.2	0	111.8	0
5	non-coated	97	41.17	0.51	71.7	0.22	362.5	0.21
6	newsprint	109.5	41.20	2.48	71.7	1.09	367.1	1.03
7	super calendered	95	80.25	0.58	136.9	0.25	698.2	0.24
8	non-coated	98	31.03	0	54.8	0	275.9	0
9	tracing plastic	89	6.5	0	12.5	0	69.1	0
10	tracing paper	130	2.92	0	5.7	0	33.5	0
11	super calendered	84	23.84	0	43.0	0	217.4	0
12	non-coated	65	62.17	0.65	108.4	0.28	547.3	0.28
13	coated	89	70.67	0.84	121.2	0.36	613.7	0.34
14	non-coated	122	25.33	0	44.7	0	224.5	0
15	non-coated	126	37.27	0.28	64.6	0.12	328.3	0.12
16	non-coated	99	32.83	0.18	57.7	0.08	292.1	0.77
17	non-coated	124	49.66	0.24	85.4	0.10	431.5	0.10
18	non-coated	104	32.4	0	56.9	0	285.0	0
19	tracing plastic	110	7.52	0	14.2	0	75.6	0
20	non-coated	293	53.01	0.49	89.4	0.20	455.2	0.20
21	coated	93	75.17	0.54	128.8	0.23	658.3	0.22
22	super calendered	130	55.25	0.48	94.6	0.20	481.6	0.19

where w is the spatial frequency. Equation (1) leads to

$$k_p = \frac{5.4}{S}, \tag{2}$$

since $\mathrm{MTF}(1/k_p) = 0.5$. The k_p values from the MC simulations are given by the the inverse frequency at which $\mathrm{MTF} = 0.5$ (Fig. 1(c)).

3. Results

The MC simulated LSF for the opaque pad case is shown in Fig. 1(b). Only one translucent paper (Sample 2) and an uncoated paper (sample 17) are shown for clarity. The LSF of translucent samples is significantly different for the case with $g = 0$ than for the case with $g = 0.8$. The shape of the MTF in Fig. 1(c) also differs but the k_p values are close. For these samples, the Arney MTF model (Eq. (1)) gives MTFs similar to the MC simulated MTF with $g = 0$. For non translucent samples, Eq. (1) and MC simulations differ at larger frequencies, but the k_p values are close. Hence the k_p metric does not reflect the different MTFs obtained with the different models, nor does it reflect the effect of the asymmetry factor on the MTF.

On the other hand, the effect of anisotropic single scattering on the MTF can be assessed by comparing the MTF curves at different values of the asymmetry factor. As shown in Fig. 1(c), forward scattering decreases the MTF at all frequencies above $1/k_p$. This is in line with

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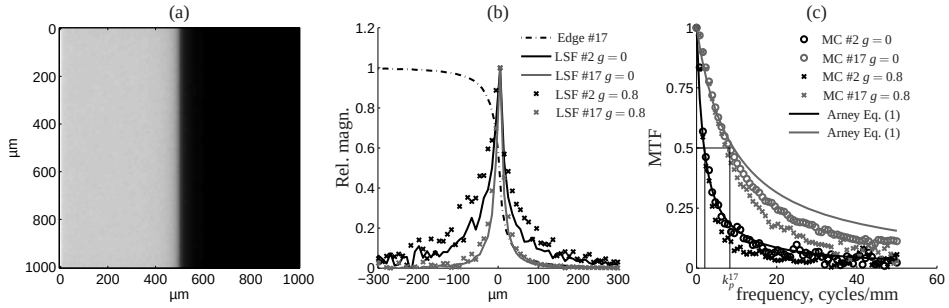


Fig. 1. (a) Simulated reflectance image. (b) Averaged edge response and its derivative, the LSF, here shown for sample 2 and 17. (c) Fourier transform of the LSF gives the MTF from which k_p can be obtained.

the broadening of the point spread function with increasing forward scattering observed by Neuman et al. [16].

Figure (2) shows the MC simulated k_p versus the k_p predicted by the Arney model for all 22 samples. Figure 2(a) shows a single sample over a black background and Fig. 2(b) shows an opaque pad of identical samples. The Arney value is close to MC value for single sheets, and for opaque pads of sheets when k_p is low. The asymmetry factor has a negligible impact, except for the single sheet samples with low S (and thus larger k_p). Figure 3 shows the MC simulated k_p versus the measured k_p. The two models give similar predictions of k_p for non translucent samples (lower S and lower k_p). For the opaque pads of translucent samples, the MC simulations predict larger k_p, which better correlate with the measured k_p (Fig. 3).

For opaque samples, the correlation between MC simulated k_p and measured k_p is high. However, the MC simulations underestimate k_p, i.e. the width of the MTF at half maximum value, meaning that the models predict less lateral scattering than what is actually measured. Anisotropic scattering does not significantly increase k_p, and thus cannot explain the narrower MTF measured from paper.

4. Discussion and conclusions

The MC simulations show that the asymmetry factor has a significant impact on the shape of the MTF. Forward single scattering leads to more lateral scattering of the light. The Arney MTF model gives significantly different MTF curves, but the MTF metric k_p is close for MC and Arney models. This means that k_p as single metric of the MTF is inappropriate since it does not reflect the different MTF obtained with different asymmetry factors and with the Arney MTF model. Using k_p may lead to the wrong conclusion that the asymmetry factor does not affect the MTF.

The measurement of the MTF of paper is a difficult task. The knife-edge method used by Arney et. al. is experimentally efficient but the derivation of the LSF from the edge response increases noise in the knife-edge response. Ukishima recently developed an improved method to measure the MTF of paper [18] that should be used in future work.

The measured MTFs were not reported in [10] and it is therefore not possible to test whether models with forward single scattering would predict better the measured MTF of paper. On the other hand, it is clear that the MC simulations overestimate the width of the MTF at half maximum and thus cannot explain the large lateral scattering in paper. Surface reflection due

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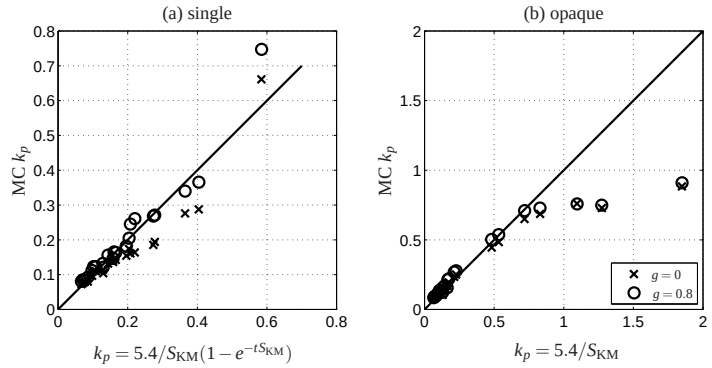


Fig. 2. Simulated k_p versus k_p obtained from Eq. (2), for single sheet (left) and opaque pad of sheets (right), and two values of the asymmetry factor g. The Arney value is close to the MC value for single sheets, and for opaque pads of sheets when k_p is low. The asymmetry factor has a negligible impact, except for the single sheet samples with low S (and thus larger k_p).

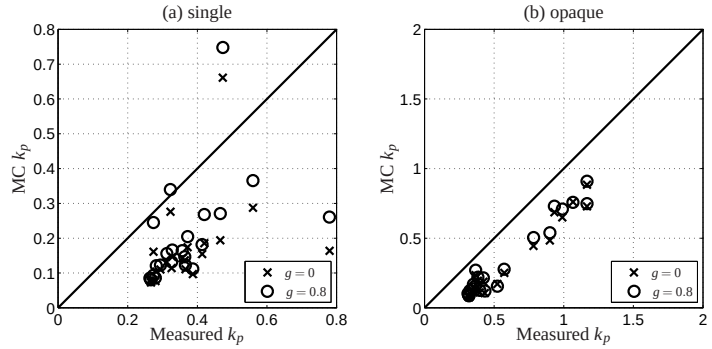


Fig. 3. Simulated k_p versus measured k_p. Bold line shows a one to one relationship. The MC simulations underestimate k_p.

to refractive index mismatch at the boundary between the air and the layer is not included in the simulations. However, surface reflection will widen the simulated MTF since gloss does not result in lateral scattering and since internal reflection at the top surface would require a larger determined scattering coefficient to compensate for the reflectance decrease. This increase of σ_s will then lead to less lateral scattering within the layer. Although single forward scattering leads to more lateral scattering than isotropic single scattering, this effect does not explain the narrower MTF measured. Directional inhomogeneity in uncoated papers and other fibrous materials such as tissue calls for a radiative transfer model with both the single scattering phase function and scattering distances being dependent on the absolute direction within the material. MC modeling of the lateral light scattering in fiber networks is ongoing to address this.

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